MLOps Specialization Bootcamp

Module 0

Onboarding and Setup

Week 0: Get your machine ready for 14 weeks of hands-on work





Daniel Islam
Instructor, InnovatiCS MLOps Specialization

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What you will be able to do



1. Work in a Linux environment: WSL2 + Ubuntu, or your macOS or Linux terminal
2. Edit and run code inside Linux with VS Code
3. Use the terminal for files, folders, pipes, `grep`, and environment variables
4. Run a safe AWS account: budget alert, IAM user, AWS CLI configured
5. Prove it all works with `scripts/verify_setup.sh`

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01

Welcome and course map

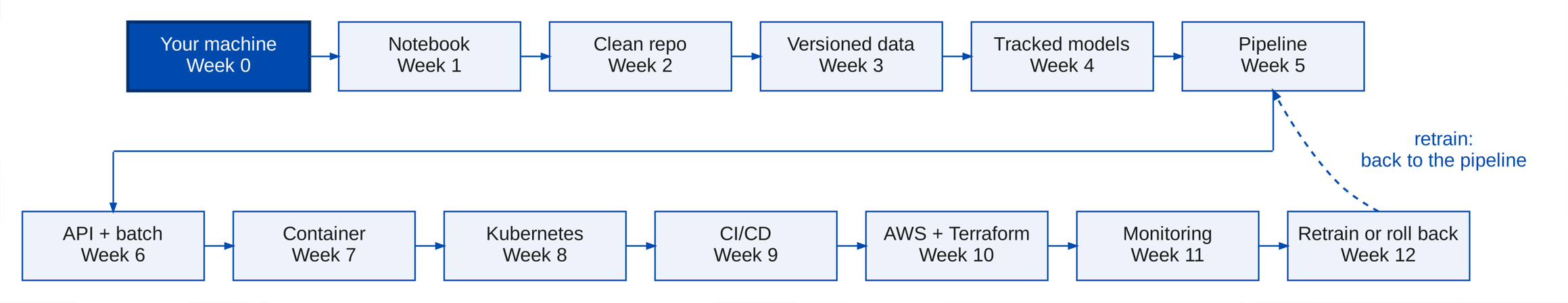
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One model, 14 weeks

- This course teaches how to **run** ML. It does not teach better models
- One project all course: Diamond Price Prediction, in `diamond-mlops/`
- Each week adds one practice: tests, data versions, tracking, an API, a container
- Week 0 adds nothing. You only prove you can run it
- Today the project is one notebook, one CSV file and 3 scripts

Where we are



Week 0 makes sure every machine in the class can run the same project.

How a week works, and how you are graded

Part	What happens	Grade
Before the session	Short reading or videos	none
Live session	About 3.5 hours: concepts, demo, practical start	none
Practical	Build this week's step on the diamond project	40%
Quiz	10 questions, opens at the end of the session	10%
Checkpoints and capstone	2 checkpoints, then your own project in Weeks 13 and 14	20% + 30%

02

Your environment

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"Works on my machine"

What it is

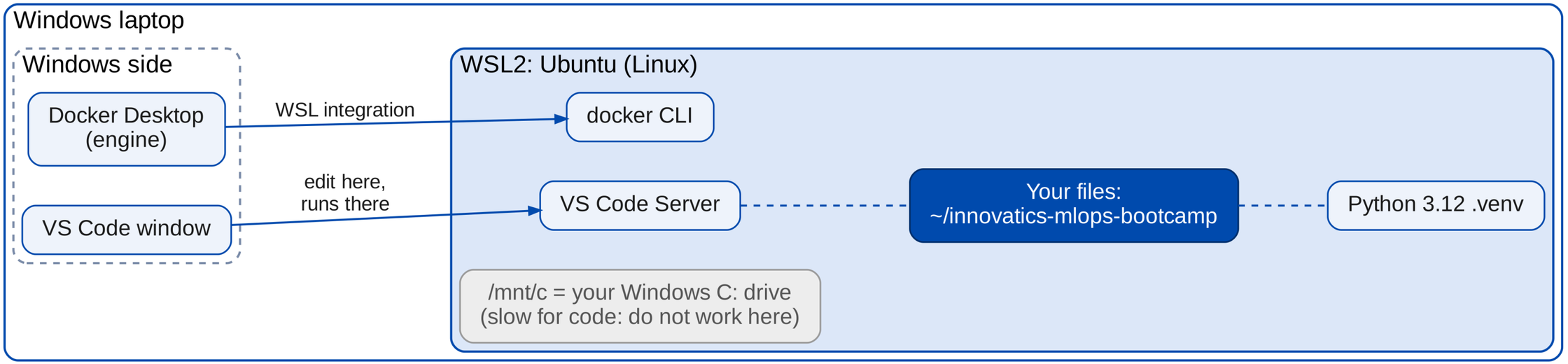
- Code runs for one person and fails for another
- The machines differ: OS, paths, Python and package versions, line endings

What we do about it

- The whole class uses one standard environment: Linux
- MLOps tools assume Linux, and production servers and containers run it

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Where does my code run?



Keep your work under `~` in Ubuntu. Our test install took over 15 minutes under `/mnt/c` and under 1 minute in `~`.

One Python, one environment per project, locked

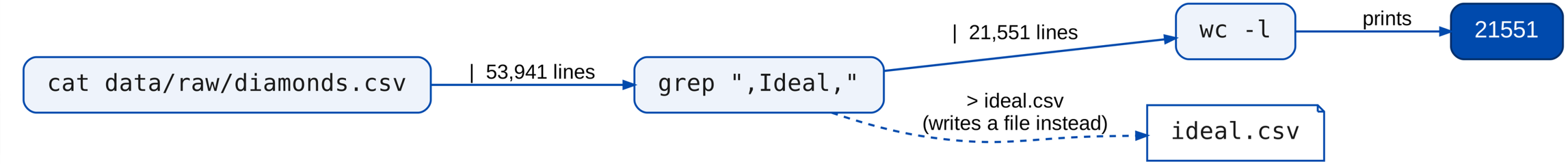
```
python3.12 -m venv .venv
.venv/bin/python -m pip install --require-hashes -r requirements.txt
.venv/bin/python -m pip check
```

README.md (lines 16-18) at tag m00

`--require-hashes` installs only the exact locked files. `pip check` should print "No broken requirements found."

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The terminal: the pipe



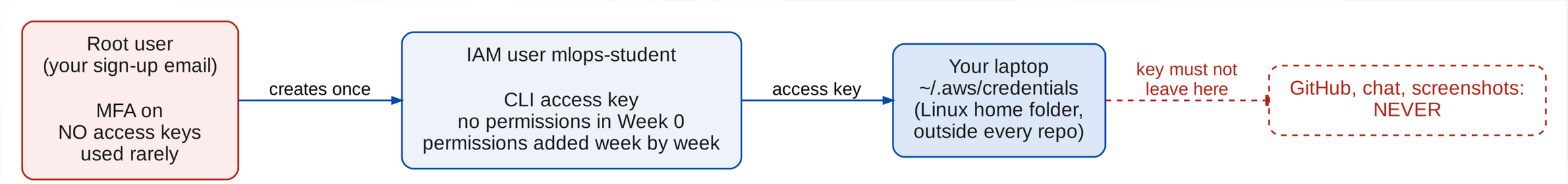
`|` sends one command's output into the next. `>` writes it to a file and replaces the file. `>>` adds to the end.

03

AWS safety

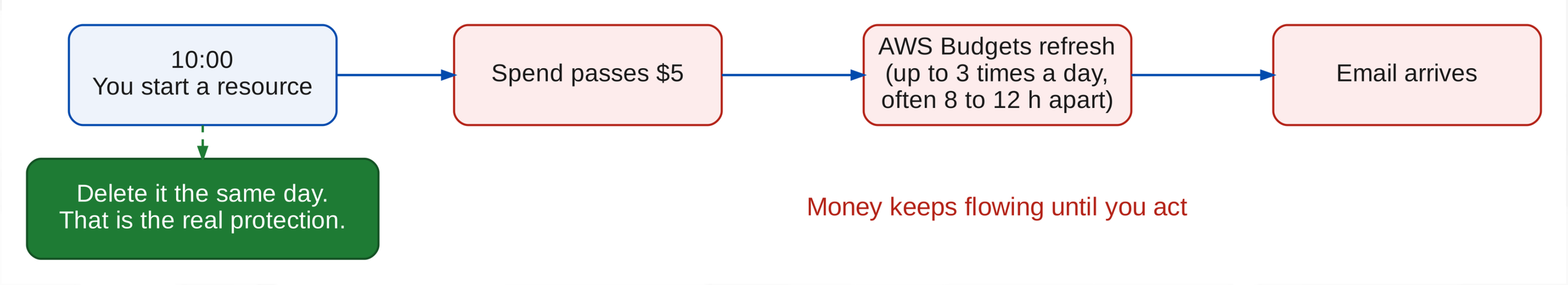
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Three layers of AWS access



Never create access keys for the AWS root user. A leaked root key gives a stranger full control.

The alert comes after the spend



A budget alert is a smoke alarm. It only sends an email. It does not stop anything.

04

Getting help

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Top 5 failures and their fixes

You see	Fix
WSL error 0x80370102 or 0x80070003	Turn on virtualization in BIOS/UEFI and the Virtual Machine Platform feature
Install looks frozen for many minutes	Clone again under `~`, not under `/mnt/c`
`python3.12: command not found`, or `unzip` missing	Use Ubuntu 24.04 (`wsl --install -d Ubuntu-24.04`). Then `sudo apt install -y python3-venv unzip`
`FAIL Docker`	Start Docker Desktop. Turn on WSL integration for Ubuntu-24.04
`FAIL AWS identity`	You used root or SSO keys. Delete root keys; configure `mlops-student`

How to ask for help

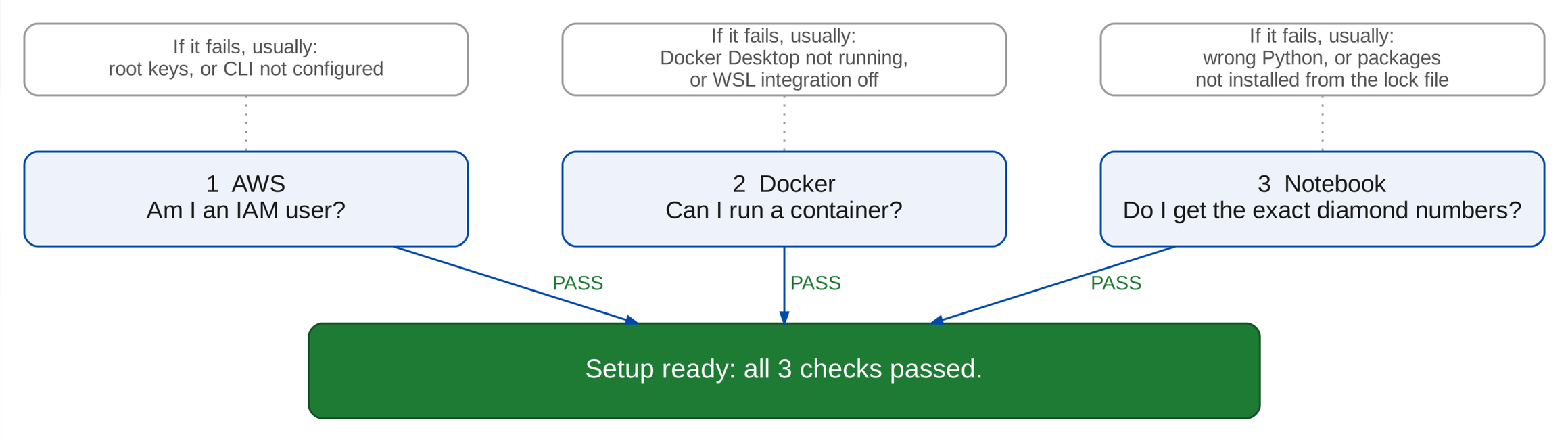
- First read the last lines of the error, then search the course FAQ and threads
- Post what you ran, the full error **as text**, and what you already tried
- Say your setup: Windows with WSL2, macOS or Linux
- One problem per thread. Post your fix when you find it
- **Never** paste access keys, passwords or `get-caller-identity` output

05

Run the check

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Three checks, one green line



Run `bash scripts/verify_setup.sh` from `diamond-mlops/`. It tries all 3 checks, then reports.

Run it now

```
if (( failures > 0 )); then
    echo "Setup incomplete: $failures of 3 checks failed." >&2
    exit 1
fi
echo 'Setup ready: all 3 checks passed.'
```

scripts/verify_setup.sh (lines 42-46) at tag m00

Type `bash scripts/verify_setup.sh`. Paste the lines from `MAE:` to the end as your Practical 0 submission.

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THANK YOU



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